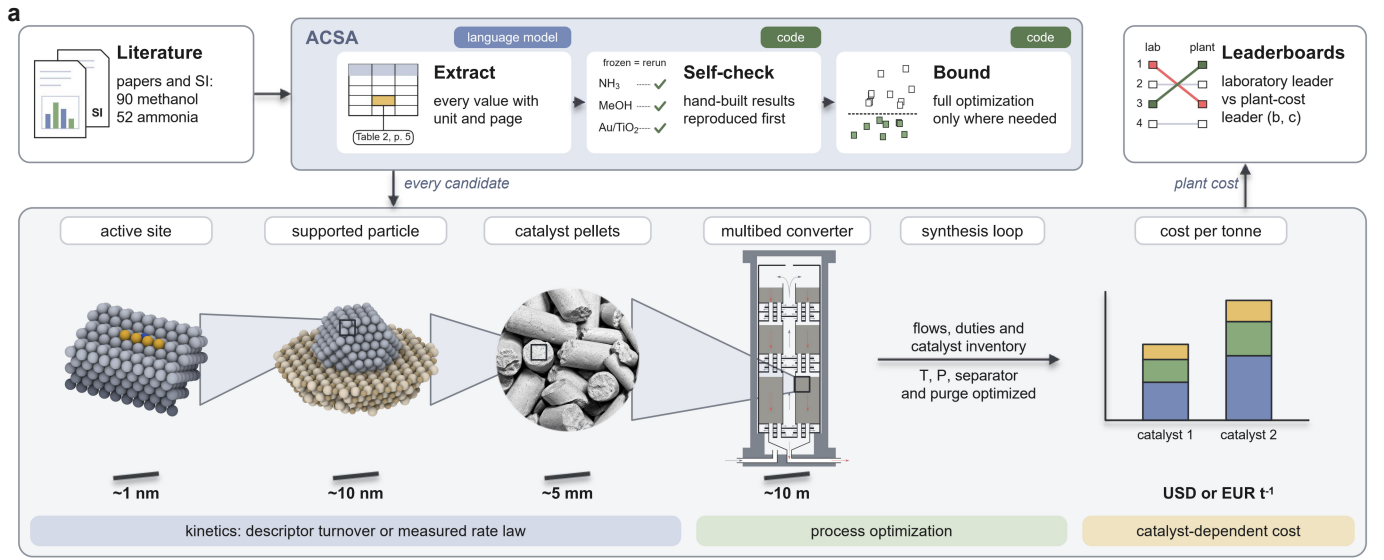
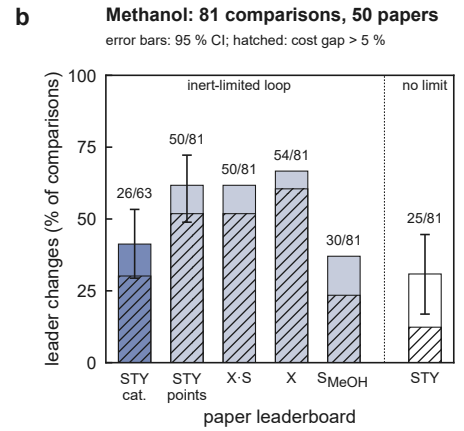
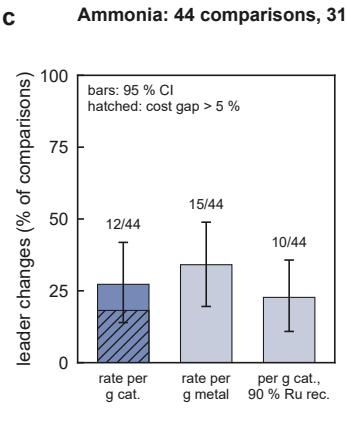
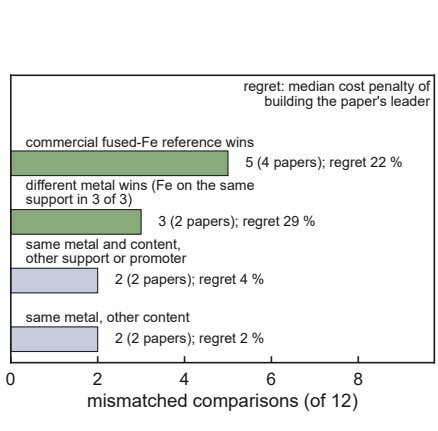
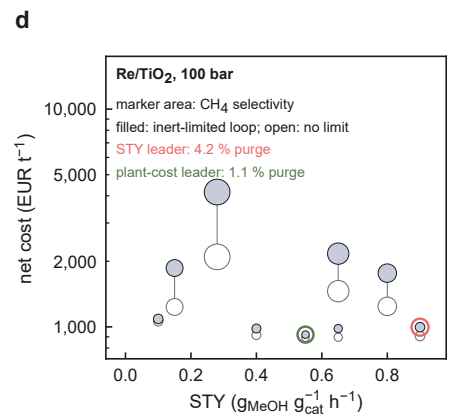
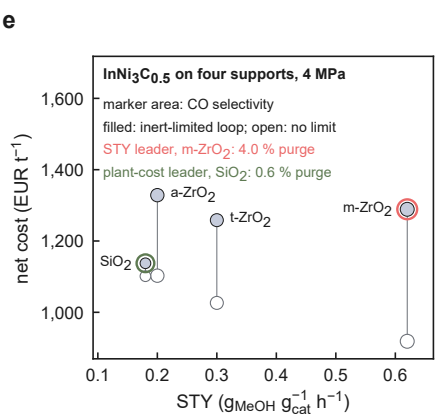

a
Literature
papers and SI:
90 methanol
52 ammonia
SI
ACSA
language model
code
code
Extract
every value with
unit and page
Table 2, p. 5
frozen = rerun
NH3
MeOH
Au/TiO2
Self-check
hand-built results
reproduced first
Bound
full optimization
only where needed
lab
plant
Leaderboards
laboratory leader
vs plant-cost
leader (b, c)
every candidate
plant cost
active site
supported particle
catalyst pellets
multibed converter
synthesis loop
cost per tonne
flows, duties and
catalyst inventory
T, P, separator
and purge optimized
catalyst 1
catalyst 2
~1 nm
~10 nm
~5 mm
~10 m
USD or EUR t-1
kinetics: descriptor turnover or measured rate law
process optimization
catalyst-dependent cost

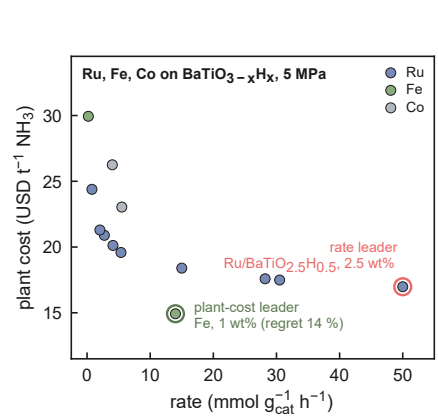

b
Methanol: 81 comparisons, 50 papers
error bars: 95 % CI; hatched: cost gap > 5 %
inert-limited loop
no limit
leader changes (% of comparisons)
26/63
50/81
50/81
54/81
30/81
25/81
STY cat.
STY points
X·S
X
SMeOH
STY
paper leaderboard


c
Ammonia: 44 comparisons, 31 papers
bars: 95 % CI
hatched: cost gap > 5 %
leader changes (% of comparisons)
12/44
15/44
10/44
rate per g cat.
rate per g metal
per g cat., 90 % Ru rec.


regret: median cost penalty of
building the paper's leader
commercial fused-Fe reference wins
5 (4 papers); regret 22 %
different metal wins (Fe on the same
support in 3 of 3)
3 (2 papers); regret 29 %
same metal and content,
other support or promoter
2 (2 papers); regret 4 %
same metal, other content
2 (2 papers); regret 2 %
mismatched comparisons (of 12)


d
Re/TiO2, 100 bar
marker area: CH4 selectivity
filled: inert-limited loop; open: no limit
STY leader: 4.2 % purge
plant-cost leader: 1.1 % purge
net cost (EUR t-1)
STY (gMeOH gcat-1 h-1)


e
InNi3C0.5 on four supports, 4 MPa
marker area: CO selectivity
filled: inert-limited loop; open: no limit
STY leader, m-ZrO2: 4.0 % purge
plant-cost leader, SiO2: 0.6 % purge
a-ZrO2
t-ZrO2
m-ZrO2
SiO2
net cost (EUR t-1)
STY (gMeOH gcat-1 h-1)


f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
Ru
Fe
Co
rate leader
Ru/BaTiO2.5H0.5, 2.5 wt%
plant-cost leader
Fe, 1 wt% (regret 14 %)
plant cost (USD t-1 NH3)
rate (mmol gcat-1 h-1)